Mahmoud Alyosify

Edge & Efficient AI Engineer

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SUMMARY

Engineer focused on making AI models **fast and cheap enough to run**. My MSc thesis cut 47.6% of LLM output tokens and made answers 19–68% faster, with no answer cut off. Hands-on with ONNX export, 4-bit fine-tuning (QLoRA), local models on a 4 GB GPU, and C/C++. MSc in Artificial Intelligence, Queen's University, Canada (GPA 4.30 / 4.30).

SKILLS

Efficiency: inference optimisation, latency profiling, quantisation (4-bit, QLoRA), caching
Runtimes: ONNX, Ollama, FAISS, FastAPI, Docker
Machine learning: PyTorch, Hugging Face Transformers, TensorFlow
Programming: Python, C, C++

PROJECTS

MinimalLM — Master's thesis (Python, PyTorch, Hugging Face Transformers) 2026

- A training-free method that makes open LLMs (Mistral-7B, Llama-3.1-8B, Qwen2.5-3B) give shorter answers that are still complete: **47.6% fewer tokens, 19–68% faster, no answer cut off** (0 of 990).
- Cost per token stayed within 0.8% of the original, so the time saved is real; GPU time in fixed batches fell 5–55%. Measured on an NVIDIA A40.

SimCLR-Vision-SSL — self-supervised image model (PyTorch, ONNX, FAISS) 2026

- Exported the model to **ONNX** and served image search through a **FAISS** index, ready to deploy.

Horus-OSINT — fine-tuned LLM on AWS (Llama-3-8B, QLoRA, AWS) 2026

- Fine-tuned Meta-Llama-3-8B with **4-bit QLoRA** on 159,826 records and served it self-hosted on AWS, keeping the model and data in one environment.

EXPERIENCE

AI Engineer Intern — Injaz Tech (Injaz Group), Cairo Oct 2026 – Present

- Ran the assistant on a local **3B model** (Ollama) on a 4 GB GPU; a response cache cut repeat answers from **34.7 s to 1.6 s**.
- Built a RAG assistant that answers staff questions from company documents, using **LangChain, LangGraph, FAISS** and **FastAPI**.

Systems Engineer (Military Service) — National Company for Roads Building & Development, Cairo
Feb 2024 – Mar 2025

- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles**.
- Debugged live transportation software and supported legacy applications to keep downtime to a minimum.

Part-Time Machine Learning Instructor — National Telecommunication Institute (NTI), Egypt Jul 2025 – Present

- Teach **90-hour machine learning bootcamps** and AI for Business modules in NTI's Summer Internships Program.

Online Instructor — Udemy & YouTube (Arabic courses), Remote 2020 – Present

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

EDUCATION

MSc in Artificial Intelligence — Queen's University, Kingston, Canada Sep 2025 – Oct 2026

- GPA **4.30 / 4.30** · Completed October 2026 · Digilians Presidential Scholarship
- Thesis: MinimalLM — shorter LLM answers that stay complete (supervisor: Dr. Rahatara Ferdousi)

BSc in Computer & Information Science (Bioinformatics) — Assiut University, Egypt Sep 2019 – Jul 2023

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+

CERTIFICATIONS & AWARDS

Deep Learning Specialization (DeepLearning.AI, 2026) · Machine Learning Specialization (DeepLearning.AI) · Machine Learning Internship, ITIDA / Egypt Makes Electronics (200 hours) · 1st Place, Smart Cities Hackathon 2022 (77 university teams)

LANGUAGES

Arabic (native) · English (professional working proficiency)